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Note

The experimental verification on Einstein's theoretical explanation of Brownian motion was done by Jean Perrin in the year 1908. The Einstein's explanation on Brownian motion and Perrin experiment was of great importance in physics because it provided direct evidence of reality of atoms and molecules.

SUMMARY

- Kinetic theory explains the microscopic origin of macroscopic parameters like temperature, pressure.
- The pressure exerted on the walls of gas container is due to the momentum imparted by the gas molecules on the walls.
- The pressure $P = \frac{1}{3} nm\overline{v^2}$. The pressure is directly proportional to the number density, mass of molecule and mean square speed.
- The temperature of a gas is a measure of the average translational kinetic energy per molecule of the gas. The average kinetic energy per molecule is directly proportional to absolute temperature of gas and independent of nature of molecules.
- The pressure is also equal to 2/3 of internal energy per unit volume.
- The rms speed of gas molecules = $v_{rms} = \sqrt{\frac{3kT}{m}} = 1.73\sqrt{\frac{kT}{m}}$
- The average speed of gas molecules $\bar{v} = \sqrt{\frac{8kT}{\pi m}} = 1.60\sqrt{\frac{kT}{m}}$
- The most probable speed of gas molecules $v_{mp} = \sqrt{\frac{2kT}{m}} = 1.41\sqrt{\frac{kT}{m}}$
- Among the speeds v_{rms} is the largest and v_{mp} is the least

$$v_{rms} > v > v_{mp}$$

- The number of gas molecules in the range of speed v to v+dv is given by Maxwell-Boltzmann distribution

$$N_v dv = 4\pi N\left(\frac{m}{2\pi kT}\right)^{\frac{3}{2}} v^2 e^{-\frac{mv^2}{2kT}} dv$$

- The minimum number of independent coordinates needed to specify the position and configuration of a thermodynamical system in space is called the degrees of freedom of the system. If a sample of gas has N molecules, then the total degrees of freedom f = 3N. If there are q number of constraints then total degrees of freedom f = 3N-q.
- For a monoatomic molecule, f = 3

  For a diatomic molecule (at normal temperature), f = 5